

Enabling Visually Aware Conversational Data Visualization Through LLM Augmentation

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Abstract

We present an augmentation method that makes Large Language Models (LLMs) both data aware and visually aware. Unaugmented LLMs can provide high-quality information about the broad context of a visualization, but are unaware of the visual content and thus cannot provide accurate, visualization-specific answers. We address this limitation by providing LLMs with structured meta-data generated from a combination of extracted visual information and textual descriptions. Our LLM-agnostic approach preprocesses the visualization to extract features from it using a vision model, combines them with textual information about the data, and generates a compact JSON file that then augments the LLM during user interactions. This highly structured file provides the LLM with the necessary multimodal context without requiring any fine-tuning or costly multimodal prompts, and applies to any existing pre-rendered visualization paired with descriptive text. We demonstrate our method using geospatial datasets from the Science On a Sphere project. A user study confirms our system's accuracy and appeal to users.

CCS Concepts

• **Human-centered computing** → *Empirical studies in visualization*.

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Interactivity, Accessibility, Visualization, Learning, Language Models, Multi-platform

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1 Introduction

Visualizations convey complex phenomena, yet many viewers struggle to interpret what they see without guidance [Smith et al. 2011]. Human docents can bridge this gap by explaining encodings and connecting patterns to domain knowledge, but scaling human facilitation is costly. Large Language Models (LLMs) can handle natural language questions and carry broad background knowledge, but unaugmented LLMs are not grounded in the specific visualization at hand. Meanwhile, vision-language models introduce prohibitive latency for interactive experiences [Shi et al. 2025], and incur high costs when used through API services such as OpenAI's.

We present a lightweight augmentation that makes LLMs both *context aware* and *visually aware* for *pre-rendered* visualizations. Our method preprocesses a visualization to extract salient visual cues and merges them with its textual description into a compact, structured JSON file. At runtime, this JSON primes the LLM to produce *visio-verbal* answers: responses that correctly reference visual information while leveraging the broader context. We demonstrate the approach on NOAA's Science On a Sphere (SoS) geospatial datasets and evaluate answer quality and user experience.

Our contributions are: (1) a model-agnostic augmentation pipeline that gives foundation LLMs visual awareness for existing, pre-rendered visuals; (2) a WebGPU-based, multi-device prototype virtual docent for SoS content; and (3) an empirical study showing accuracy gains over text-only and unstructured image-feature baselines, with high usability ratings.

2 Design Requirements

We identify three design requirements for a deployable “virtual docent” for visualizations. **Visual awareness:** refer to visual encodings and features (colors, glyphs, regions, patterns). **Contextual grounding:** connect encodings to domain meaning using dataset metadata and background knowledge. **Extensibility and practicality:** add new datasets by running an offline script; avoid model fine-tuning; keep latency and cost low; support web deployment across devices and languages.

3 Approach

Our pipeline has an *offline* preprocessing stage per dataset and an *online* question-answering (Q&A) stage per user query.

Offline: visual and textual summarization into a structured JSON. We sample a small number of representative frames (2–20, depending on dynamics) directly from the pre-rendered visualization prior to spherical projection. A vision model extracts salient cues from these frames (e.g., color-scale legends, color regions and their meanings, symbols and glyphs, prominent spatial patterns). In parallel, we parse the dataset’s text description (scope, period, provenance, intended interpretation). Because descriptions may omit visual encoding details, visual analysis fills those gaps.

An LLM (here GPT-4o) is then instructed to consolidate the extracted cues and description into a concise, schema-guided JSON with fields such as *Description*, *Period*, *Locations*, *VisualCues*, *ColorEncoding*,.... This step converts multimodal, redundant inputs into a compact knowledge representation that is reliable to parse at runtime and cheap to transmit. No model fine-tuning is required.

Online: conversational Q&A with structured augmentation. At runtime, the client passes user queries (text or voice) and the dataset’s JSON to the LLM. The LLM composes answers that point to visible structures (e.g., color bands, eddies, current boundaries) and explains their meaning using the dataset’s context. Responses are returned as text (and optionally synthesized speech).

Prototype system. The visualization front end is implemented in WebGPU and streams equirectangular videos as GPU textures; we support globe projection and interactive navigation on phones, PCs, and HMDs. Typical Q&A latency is within a few seconds. Language and vision model APIs are pluggable.

4 Demonstration

We illustrate the docent on SoS datasets such as *Loggerhead Sea Turtle Migration* (turtle tracks over sea surface temperature) and *NOAA Sea Surface Temperature (SST)*. The JSON encodes the palette semantics (e.g., blue → cooler, green/yellow → ~18.5–19 °C band, red → hotter), salient regions/currents, and typical patterns (e.g., migration alignment with optimal temperatures). At runtime, the docent autonomously grounds questions like “*What do the yellow areas mean here near Japan?*” by mapping color to temperature and linking the band to seasonal shifts and the Kuroshio Current.

5 Evaluation

Setup. We compared three prompting conditions on the same questions: **TextAug** (text description only), **TextImgAug** (description + unstructured visual-feature list), and **StructAug** (our JSON). Seventeen participants (mean age 26) explored one or more datasets via voice or text. We collected 125 real user questions, then generated answers for each condition and rated their accuracy in a blind review on a 0–10 scale.

Answer quality. **StructAug** significantly outperformed both **TextAug** and **TextImgAug**. Despite **TextImgAug** having more raw tokens, the structured JSON yielded higher accuracy, consistent with findings that well-organized inputs help LLMs focus on relevant facts [Bendeck and Stasko 2024].

User experience. Post-study SUS scores averaged **83.8** ($SD \approx 4.7$), above the common benchmark of 70 [Bangor et al. 2008]. Participants strongly disagreed with needing technical support (mean 1.17/5) and with requiring prior learning (1.17/5), and agreed that most people would learn to use the system quickly (4.5/5). Free-form comments highlighted naturalness (“like having a guide next to the exhibit”) and robustness to open-ended questions.

6 Limitations and Future Work

Our preprocessing samples a small set of frames; richer handling of time (e.g., feature tracking and temporal references) could improve answers about transient events. Prompt styles could be adapted to audience (e.g., children vs. experts). While we focus on geospatial visuals, the approach generalizes to other domains with pre-rendered imagery (medical, microscopy, simulations) when encodings and common patterns can be summarized.

7 Conclusion

We show that a compact, schema-guided JSON—extracted once from a visualization and its description—can make a foundation LLM visually aware at runtime, enabling grounded, conversational explanations without fine-tuning or heavy multimodal inference. On SoS datasets, the approach yields better answer quality than text-only or unstructured augmentations and is well-liked by users. This points to practical, scalable “virtual docents” that increase access and engagement for public-facing visualizations.

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